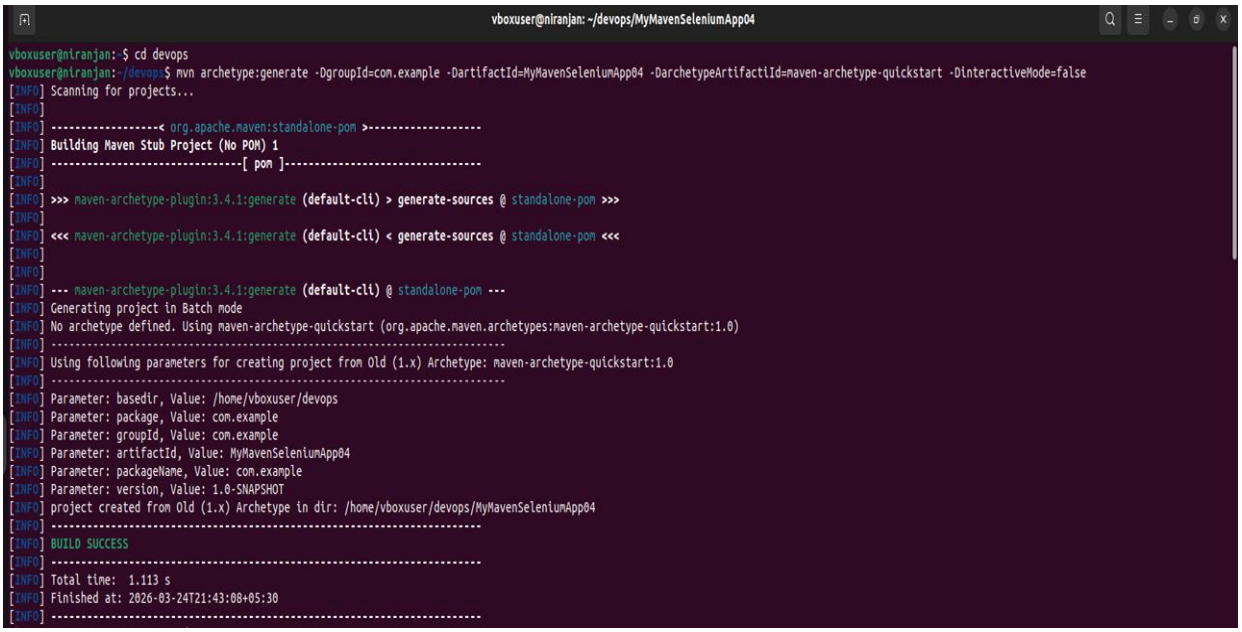
Program 6 : Working with Selenium:

Maven Project To Connect all the above 3 websites To Web Servers Using Selenium Dependency on Google Chrome.

Step 1: create a maven project on terminal by typing the command **mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp04 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false**



```
vboxuser@niranjan: ~/devops/MyMavenSeleniumApp04
vboxuser@niranjan:~$ cd devops
vboxuser@niranjan:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp04 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/vboxuser/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp04
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vboxuser/devops/MyMavenSeleniumApp04
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  1.113 s
[INFO] Finished at: 2026-03-24T21:43:08+05:30
[INFO] ----------------------------------------------------------------------------
```

Step 2 : edit the pom.xml file and add the dependencies
The pom.xml has the code

pom.xml
~/devops/MyMavenSeleniumApp04

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenSeleniumApp04</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenSeleniumApp04</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
    <dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.41.0</version>
    <scope>compile</scope>
</dependency>
  </dependencies>
  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
            <groupId>org.apache.maven.plugins</groupId>

            <artifactId>maven-jar-plugin</artifactId>
            <version>3.0.1</version>
           <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>


        </plugin>
        <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-shade-plugin</artifactId>
      <version>3.5.0</version>
      <executions>
        <execution>
          <phase>package</phase>
          <goals>
            <goal>shade</goal>
          </goals>
          <configuration>
            <transformers>
              <transformer implementation="org.apache.maven.plugins.shade.resource.ManifestResourceTransformer">
                <mainClass>com.example.App</mainClass>
              </transformer>
            </transformers>
          </configuration>
        </execution>
      </executions>
    </plugin>

</plugins>
</build>
</project>
```

XML Tab Width: 8 Ln 81, Col 11 INS

Step 3: Edit the App.java

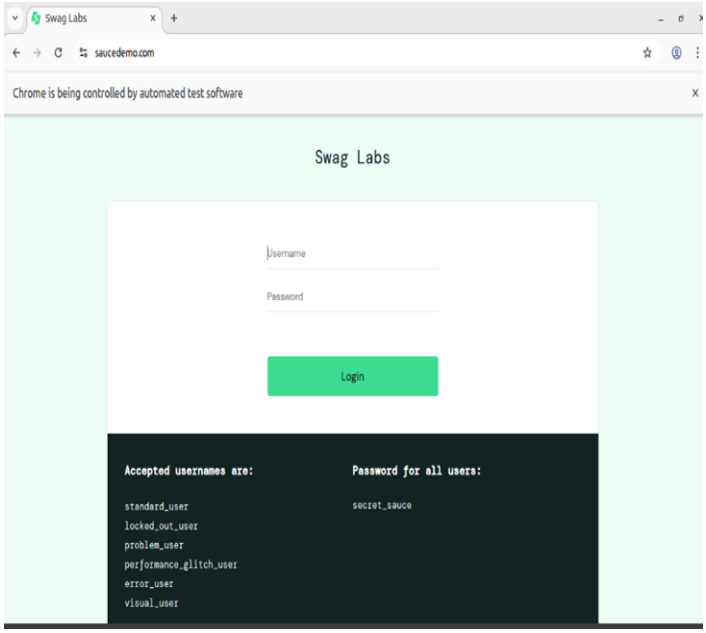
App.java
~/devops/MyMavenSeleniumApp04/src/main/java/com/example

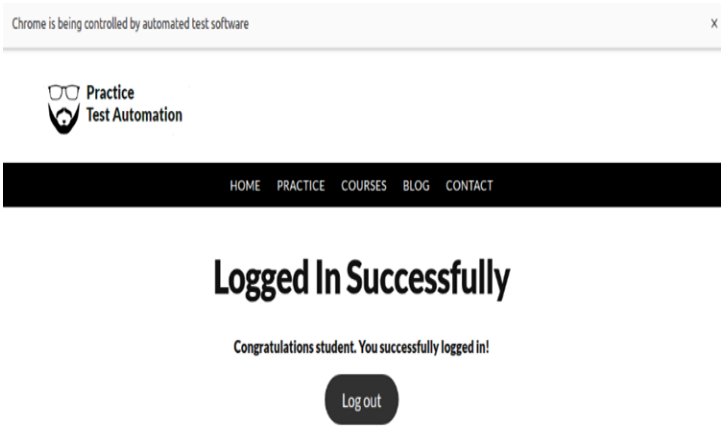
```java
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.WindowType;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;

import java.time.Duration;

public class App {

    public static void main(String[] args) throws InterruptedException {

        ChromeOptions options = new ChromeOptions();
        options.addArguments("--start-maximized"); // optional

        WebDriver driver = new ChromeDriver(options);

        driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920, 1080));

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));

        Actions actions = new Actions(driver);

        driver.get("https://www.saucedemo.com/");

        driver.findElement(By.id("user-name")).sendKeys("standard_user");
        driver.findElement(By.id("password")).sendKeys("secret_sauce");
        driver.findElement(By.id("login-button")).click();

        System.out.println("SauceDemo login successful");

        driver.switchTo().newWindow(WindowType.TAB);

        driver.get("https://automationexercise.com/products");

        driver.findElement(By.id("search_product")).sendKeys("Men Tshirt");
        driver.findElement(By.id("submit_search")).click();

        WebElement product = wait.until(
                ExpectedConditions.visibilityOfElementLocated(
                        By.cssSelector("a[data-product-id='2']")
                )
        );
```

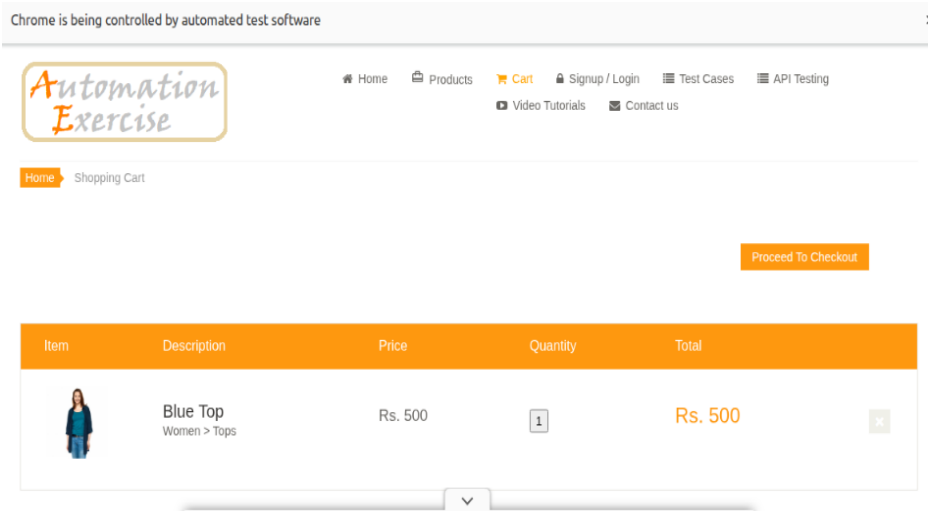
```java
        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].scrollIntoView(true);", viewCart);

        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].click();", viewCart);

        System.out.println("Automation Exercise product added to cart");

        driver.switchTo().newWindow(WindowType.TAB);

        driver.get("https://practicetestautomation.com/practice-test-login/");

        Thread.sleep(2000);

        driver.findElement(By.id("username")).sendKeys("student");
        driver.findElement(By.id("password")).sendKeys("Password123");
        driver.findElement(By.id("submit")).click();

        System.out.println("Practice Test Automation login successful");

        Thread.sleep(5000);

        driver.quit();
    }
```

Java Tab Width: 8 Ln 87, Col 2 INS

Step 4:build and run the maven application To execute type the command **<u>mvn exec:java -Dexec.mainClass="com.example.App"</u>** it opens all the 3 websites automatically and registers using username and password.

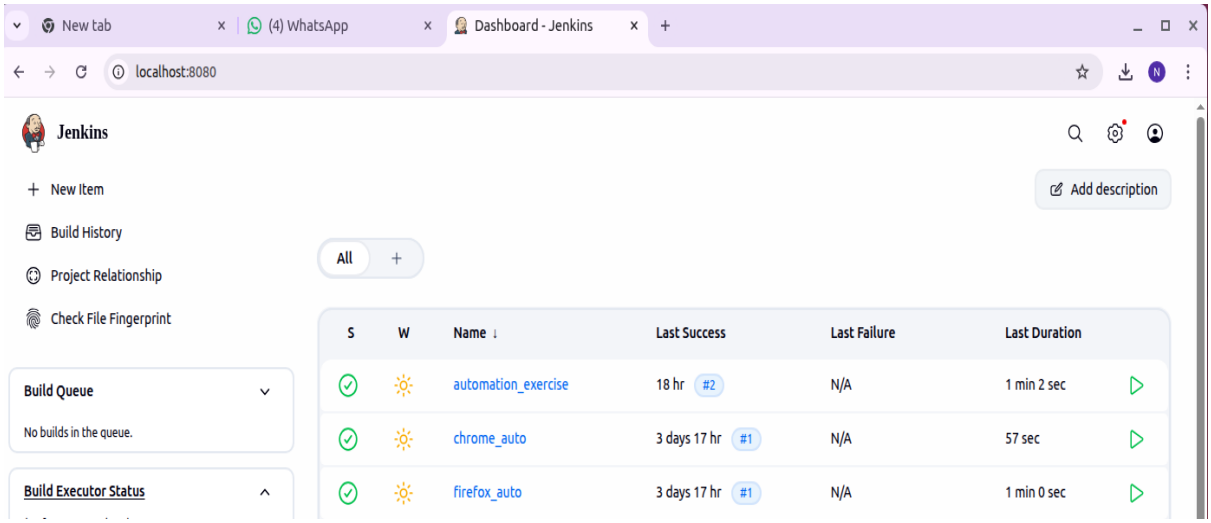




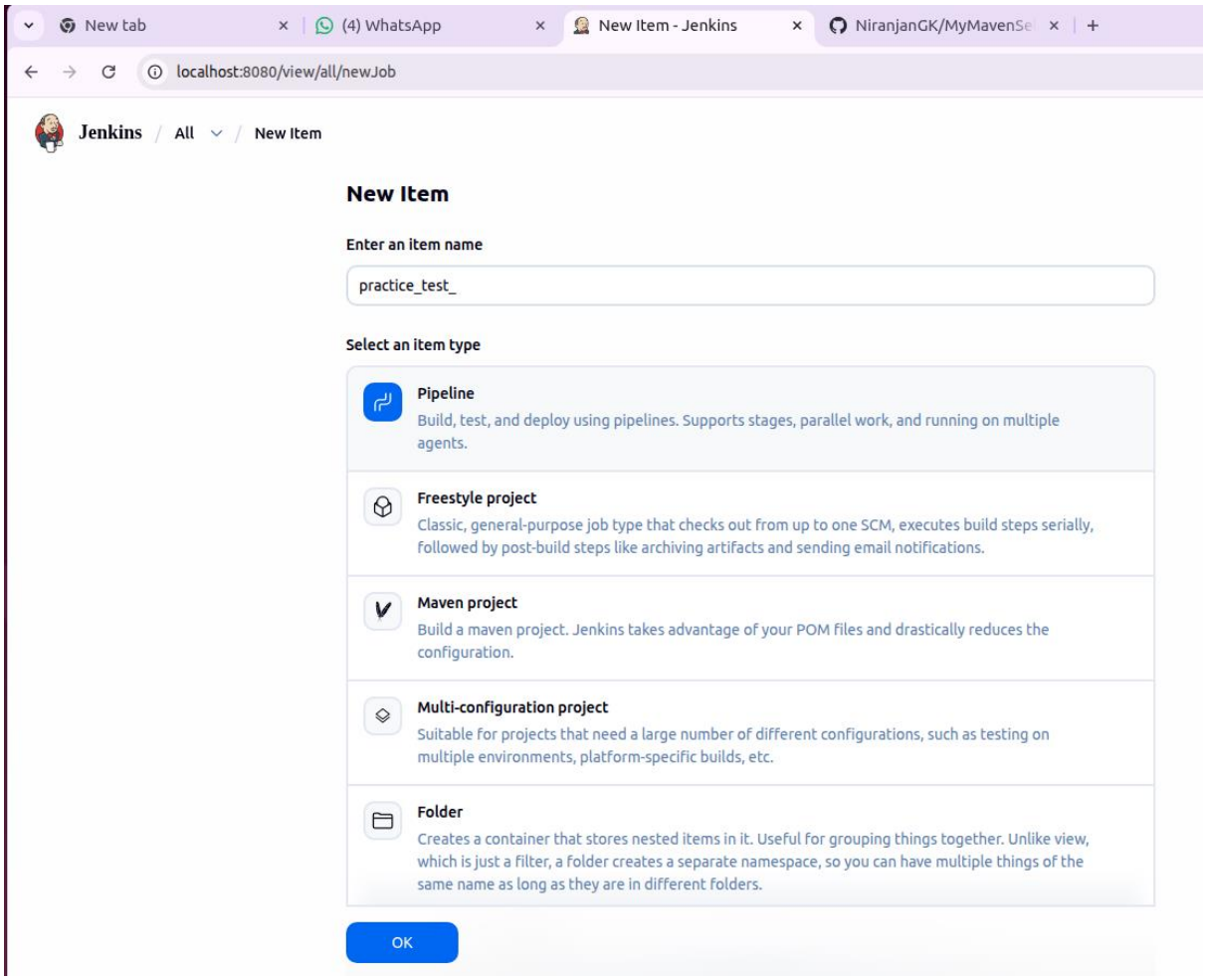


Jenkins Pipeline for MyMavenSelenium03

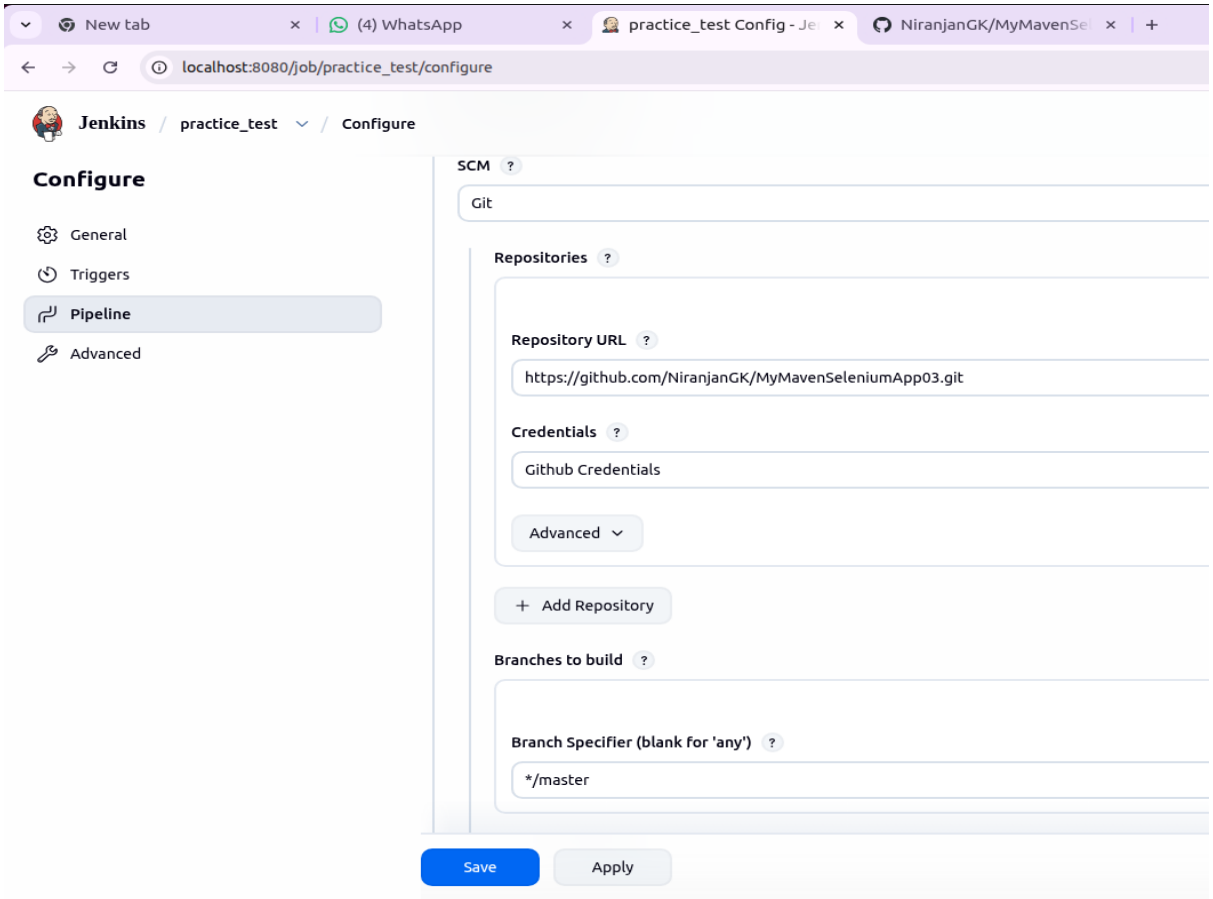
Step 1: Open the Jenkins Dashboard to create a new item



Step 2: Click on New Item to create a New Pipeline and give the item name as "practice_test" and select Item type as Pipeline and click OK



Step 3: Configure the settings set script as “Pipeline Script from SCM” and SCM as “Git”. paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenAppSelenium02 and create an new file called “Jenkinsfile”

Step 5: Paste the available code inside the Jenkinsfile

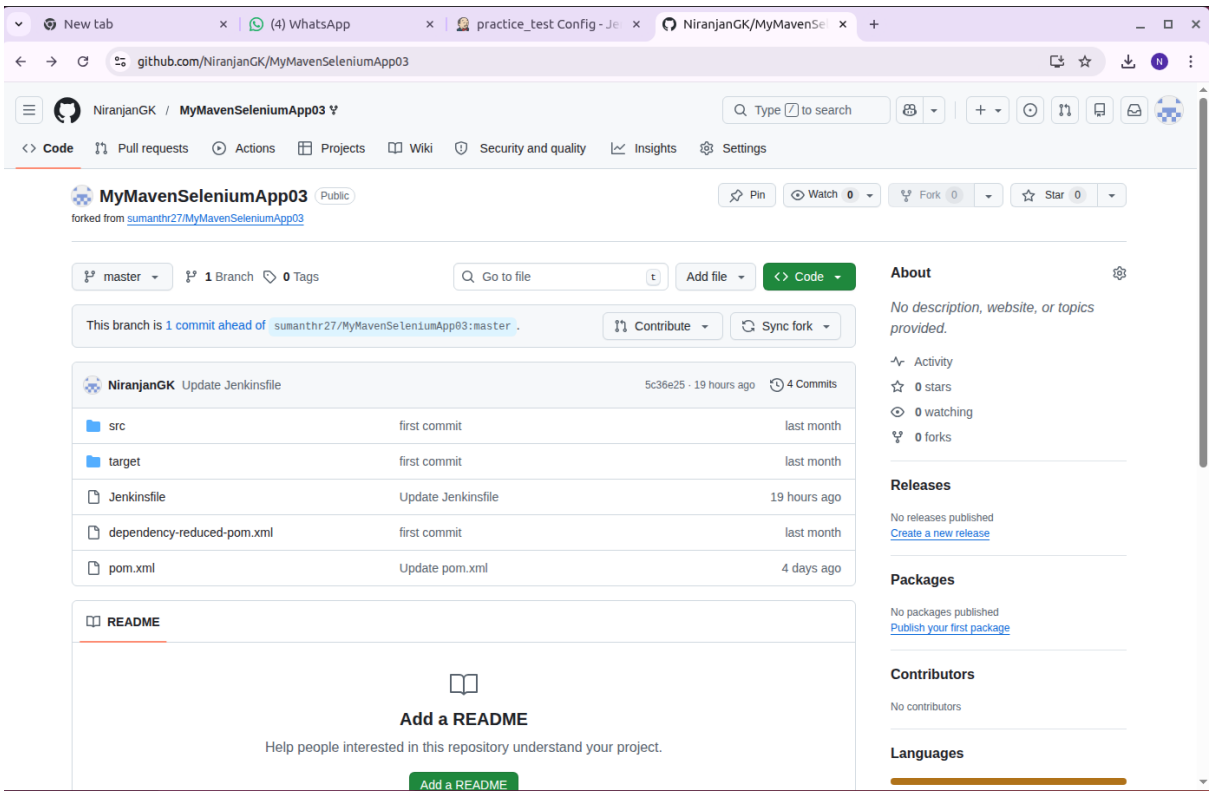
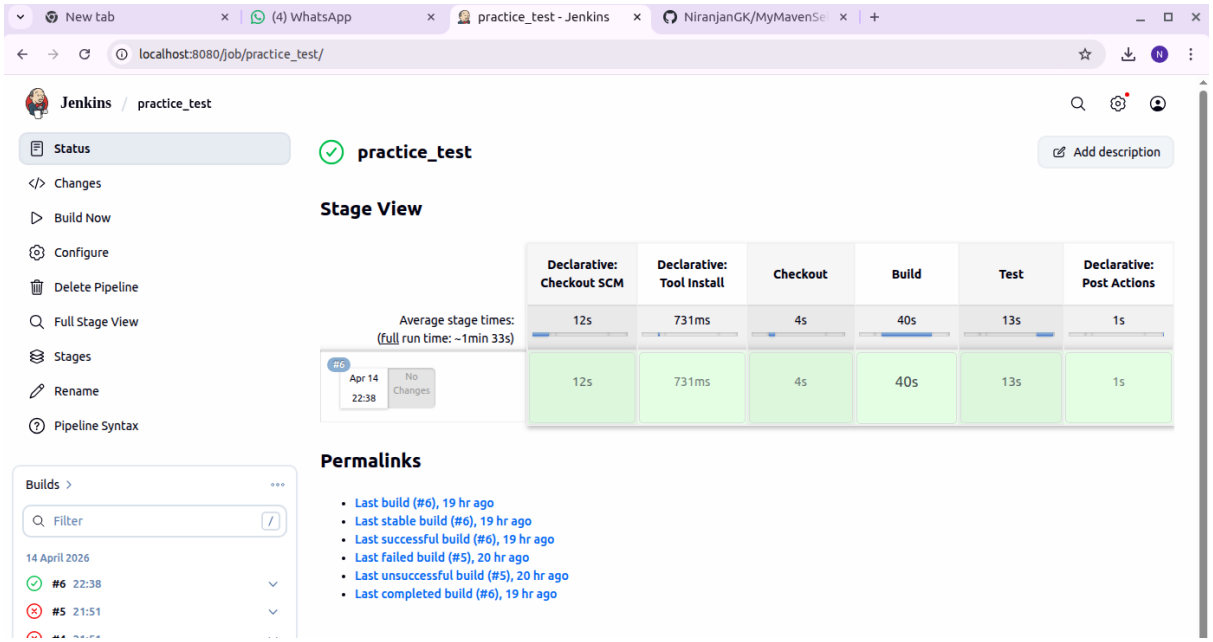
```
pipeline {
    agent any   // Run on any available agent


    tools {
        maven 'Maven'   // Must match name configured in Jenkins (Global Tool
Config)
    }


    stages {


        stage('Checkout') {
            steps {
                // Recommended if repo is configured in Jenkins
                checkout scm


                // OR use this if not configured:
                // git url:
'https://github.com/NiranjanGK/MyMavenSeleniumApp03.git', branch: 'master'
            }
        }


        stage('Build') {
            steps {
                sh 'mvn clean package'
            }
        }
```

```
        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }
    }

    post {
        always {
            echo 'Pipeline execution completed'
        }
        success {
            echo 'Build SUCCESS '
        }
        failure {
            echo 'Build FAILED '
        }
    }
}
```

Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.